

INCIDENT INVESTIGATION REPORT

Report No: INC-2024-03 | Classification: High Potential Near-Miss

Date of Incident: 14-Feb-2024 | Location: Unit 2, Pump P-101 Area

Reported by: Shift Supervisor Anil Sharma | Report Date: 18-Feb-2024

1. INCIDENT DESCRIPTION

At 14:35 on 14-Feb-2024, maintenance technician Rajan K entered the P-101 pump pit to inspect the mechanical seal (WO#9788) while a hot-work permit (HWP#2024-022) was active for welding on the adjacent cooling water line at EL+10.5 m, approximately 8 metres from the pump pit opening.

The technician's personal gas detector alarmed for LEL at 12% (above the 10% LEL threshold specified in SOP-SAF-014). The technician evacuated immediately. No injury occurred.

Gas source investigation found a small flange weep on the P-101 discharge line (not yet depressurised at the time the entry permit was issued ? procedural non-compliance).

2. IMMEDIATE CAUSE

Confined space entry was initiated before the discharge line was fully depressurised and positively isolated. The Maintenance Work Permit (MWP#2024-045) was signed by the day-shift Area Authority who had not verified isolation before handover to the incoming shift.

3. CONTRIBUTING FACTORS

- a. No SIMOPS check performed: hot-work permit HWP#2024-022 was already active in the adjacent area when CSEP was issued. SIMOPS coordinator was not informed.
- b. Shift handover gap: the Area Authority who issued the MWP was 15 minutes into handover when the permit was countersigned; isolation verification was assumed, not confirmed.
- c. Permit system siloed: CSEP and HWP are managed in separate paper-based systems with no cross-check mechanism.

4. ROOT CAUSE

Absence of an integrated permit-to-work system that enforces SIMOPS checks and prevents incompatible permits from being active simultaneously in overlapping geographic zones.

5. CORRECTIVE ACTIONS (< 7 days)

CA-01: Revoke HWP#2024-022 immediately; reinstate only after SIMOPS review. [DONE]

CA-02: Conduct toolbox talk on SIMOPS requirements with all shift teams. [DONE 16-Feb]

CA-03: Add mandatory SIMOPS coordinator sign-off field to all CSEP forms. [DONE 20-Feb]

6. PREVENTIVE ACTIONS (Long-term)

PA-01: Implement digital PTW system with geofenced SIMOPS conflict detection.

Owner: Plant Manager. Target: Q3 2024.

PA-02: Integrate CMMS work orders with PTW system to flag concurrent high-risk activities.

Owner: IT / Maintenance Dept. Target: Q4 2024.

PA-03: Annual SIMOPS competency assessment for Area Authorities.

Owner: Safety Dept. Target: Ongoing from Mar-2024.

7. LESSONS LEARNED

? Data present, action absent: gas detector readings and permit logs both existed ?

what was missing was an intelligence layer that connected them before entry.

? Simultaneous paper-based CSEP and HWP systems cannot prevent SIMOPS conflicts.

? OISD-GDN-192 SIMOPS requirements are not being enforced at permit issuance stage.

8. REGULATORY REFERENCE

OISD-GDN-192 Section 5.1: Simultaneous Operations must be identified and controlled.

Factories Act 1948 Sec 36A: Safety officers must certify hazardous process safety systems.

SOP-SAF-014 Section 5: Mandatory SIMOPS declaration before confined space entry.